

J/ApJS/249/18
variable stars

The ZTF catalog of periodic
(Chen+, 2020)

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The Zwicky Transient Facility catalog of
periodic variable stars.

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variable stars ;

Cepheid variable stars ; RR Lyrae
variable stars ;

Delta Scuti variable stars ; Long
period variable stars ;

Mira variable stars ; Eclipsing binary
stars ; Galaxy structure ;

Distance indicators

Abstract:

The number of known periodic variables has
grown rapidly in recent

years. Thanks to its large field of view and
faint limiting magnitude,

the Zwicky Transient Facility (ZTF) offers a
unique opportunity to

detect variable stars in the northern sky.

Here, we exploit ZTF Data

Release 2 (DR2) to search for and classify variables down to

$r \sim 20.6$ mag. We classify 781,602 periodic variables into 11 main types

using an improved classification method.

Comparison with previously

published catalogs shows that 621,702 objects (79.5%) are newly

discovered or newly classified, including ~ 700 Cepheids, ~ 5000 RR

Lyrae stars, ~ 15000 δ Scuti variables, $\sim 350,000$ eclipsing

binaries, $\sim 100,000$ long-period variables, and about 150,000 rotational

variables. The typical misclassification rate and period accuracy are

on the order of 2% and 99%, respectively.

74% of our variables are

located at Galactic latitudes, $|b| < 10^\circ$.

This large sample of

Cepheids, RR Lyrae, δ Scuti stars, and contact (EW-type)

eclipsing binaries is helpful to investigate the Galaxy's disk

structure and evolution with an improved completeness, areal coverage,

and age resolution. Specifically, the northern warp and the disk's

edge at distances of 15–20 kpc are significantly better covered than

previously. Among rotational variables, RS Canum Venaticorum and BY

Draconis-type variables can be separated

easily. Our knowledge of stellar chromospheric activity would benefit greatly from a statistical analysis of these types of variables.

Description:

The Zwicky Transient Facility (ZTF) is a 48-inch Schmidt telescope at the Palomar Observatory with a 47deg^2^ field of view.

ZTF DR2 contains data acquired between 2018 March and 2019 June, covering a time span of around 470 days. The photometry is provided in the g and r bands, with a uniform exposure time of 30s per observation.

ZTF DR2 includes more than 1 billion stars, about half of which have >20 epochs of observations. For the majority of stars located in the northern Galactic plane, ZTF contains ~150 epochs of observations.

File Summary:

FileName	Lrecl	Records	Explanations
ReadMe	80	.	This file
table2.dat	228	781602	ZTF variables
catalog			

table3.dat	120	1381527	ZTF suspected
variables catalog			
ztf2g.dat	75	89697393	g band single
exposure photometry data			
			of variables
from ZTF DR2			
ztf2r.dat	75	89697393	r band single
exposure photometry data			
			of variables
from ZTF DR2			

Description of files:

ztf2g.csv = g band single exposure photometry data of variables from ZTF DR2

ztf2r.csv = r band single exposure photometry data of variables from ZTF DR2

Content: Seq, RAdeg, DEdeg, MJD, mag, e_mag, flag

See also:

II/351 : VISTA Magellanic Survey (VMC) catalog (Cioni+, 2011)

I/345 : Gaia DR2 (Gaia Collaboration, 2018)

V/76 : Chromospherically Active Binaries (Strassmeier+ 1993)

V/128 : Chromospherically Active Binaries. Third version (Eker+, 2008)

J/AJ/119/1901 : ROTSE all-sky surveys for variable stars (Akerlof+, 2000)

J/AcA/51/221 : OGLE-II. Cepheids in IC 1613 (Udalski+, 2001)

J/AJ/127/861 : RR Lyrae in Draco dSph

(Bonanos+, 2004)

J/ApJ/629/1055 : Evolution of close binary systems (Yakut+, 2005)

J/MNRAS/371/1405 : Variables in M33 (Hartman+, 2006)

J/AcA/58/163 : VI light curves of LMC classical Cepheids (Soszynski+, 2008)

J/AJ/140/328 : LMC High-amplitude delta Scuti stars (Garg+, 2010)

J/AJ/141/166 : HATNet variability survey of K and M dwarfs (Hartman+, 2011)

J/ApJ/727/53 : Red supergiant stars in the LMC. I. (Yang+, 2011)

J/ApJ/754/35 : Red supergiant stars in the SMC. II. (Yang+, 2012)

J/ApJ/765/154 : RR Lyrae stars in the Catalina Sky Survey (Drake+, 2013)

J/AJ/146/101 : LINEAR. III. Cat. of periodic variables (Palaversa+, 2013)

J/AcA/63/379 : OGLE-III var. stars in Gal. disk area (Pietrukowicz+, 2013)

J/ApJS/213/9 : Catalina Surveys periodic variable stars (Drake+, 2014)

J/MNRAS/442/2620 : VRI light curve of 5 eclipsing binaries (Zhang+, 2014)

J/AJ/152/129 : VR photometry for 914 stars in NGC 188 (Chen+, 2016)

J/A+A/591/A145 : VVV Survey outer bulge RRab stars (Gran+, 2016)

J/ApJ/832/176 : Classical Cepheids in MCs. I. LMC disk (Inno+, 2016)

J/A+A/605/A100 : VVV survey Galactic Bulge Pop. II Cepheids (Bhardwaj+, 2017)

J/MNRAS/469/3688 : CSS Periodic Variable Star

Catalogue (Drake+, 2017)

J/A+A/604/A29 : MC Type II & anomalous
Cepheids PL & PR (Groenewegen+, 2017)

J/MNRAS/472/808 : YJKs light curves of SMC
Classical Cepheids (Ripepi+, 2017)

J/ApJ/844/L4 : PS1 RRab stars for tracing
Virgo overdensity (Sesar+, 2017)

J/AJ/154/149 : LMC NIR synoptic survey. V.
NIR phot. of Miras (Yuan+, 2017)

J/A+A/619/A51 : Type II Cepheids in the
Galactic bulge (Braga+, 2018)

J/AJ/155/137 : RRLs in globulars. V. $\{\omega\}$
Cen NIR phot. (Braga+, 2018)

J/ApJ/859/140 : Multi-band magnitudes for W
UMa EB* candidates (Chen+, 2018)

J/ApJS/237/28 : WISE catalog of periodic
variable stars (Chen+, 2018)

J/AJ/156/241 : Variable stars measured by
ATLAS (Heinze+, 2018)

J/AJ/156/130 : Cepheids in M31 – PAndromeda
Cepheid sample (Kodric+, 2018)

J/ApJ/855/43 : Distant RR Lyrae stars
discovered with HiTS (Medina+, 2018)

J/AJ/155/45 : Metal-rich RRc stars
(Snedden+, 2018)

J/AcA/68/315 : OGLE Galactic Cepheids
(Udalski+, 2018)

J/ApJ/852/78 : JHKs, WISE & Spitzer data of
Galactic Cepheids (Wang+, 2018)

J/A+A/616/A175 : Red supergiant stars in the
LMC. II. (Yang+, 2018)

J/AJ/156/112 : NIR Mira period–luminosity
relations in M33 (Yuan+, 2018)

J/ApJ/870/115 : Distance modulus & Fe/H of

RRLs in w Cen (Bono+, 2019)
J/A+A/622/A60 : Gaia DR2 misclassified RR
Lyrae list (Clementini+, 2019)
J/A+A/623/A110 : Gaia DR2. Variable stars in
CMD (Gaia Collaboration+, 2019)
J/AJ/158/105 : IR phot. of field RR Lyrae
variable stars (Layden+, 2019)
J/A+A/625/A14 : Reclassification of Cepheids
in the Gaia DR2 (Ripepi+, 2019)
J/other/Sci/365.478 : Milky Way classical
Cepheids sample (Skowron+, 2019)
J/AJ/158/16 : Identification of RR Lyrae
from the DES (Stringer+, 2019)
<http://irsa.ipac.caltech.edu/Missions/ztf.html>
: ZTF on IRSA

Byte-by-byte Description of file: table2.dat

Bytes	Format	Units	Label	Explanations
1- 22	A22	---	ID	ZTF variables catalog identifier
24- 29	I6	---	Seq	[1/781604] Internal source identifier
31- 39	F9.5	deg	RAdeg	Right Ascension in decimal degrees (J2000)
41- 49	F9.5	deg	DEdeg	Declination in decimal degrees (J2000)
51- 61	F11.7	d	Per	[0.03/973.5] Period
63- 68	F6.3	---	R21	[0/71] Amplitude ratio (a_2_/a_1_)

70– 74	F5.3	---	phi21	[3.14/9.5]
Phase difference				
				({phi}_2_-2*{phi}_1_)
76– 88	F13.7	d	T0	HJD of
minimum; HJD-2400000.5				
90– 95	F6.3	mag	gmag	[11.7/22]?=0
Mean Palomar/ZTF g band magnitude				
97–102	F6.3	mag	rmag	[11.2/21.4]?=0
Mean Palomar/ZTF r band				
				magnitude
104–116	F13.7	d	Per-g	[0.02/18340]?
=0 Period determined in g band				
118–130	F13.7	d	Per-r	[0.02/13242]?
=0 Period determined in r band				
132–135	I4	---	Ng	[0/2084]
Number of detections in g band				
137–140	I4	---	Nr	[0/3300]
Number of detections in r band				
142–147	F6.3	---	R21-g	[0/96] Ratio
of a_2_/a_1_ in g band				
149–154	F6.3	---	R21-r	[0/71] Ratio
of a_2_/a_1_ in r band				
156–160	F5.3	---	phi21-g	[0/9.5] The
{phi}_2_-2*{phi}_1_ in g band				
162–166	F5.3	---	phi21-r	[0/9.5] The
{phi}_2_-2*{phi}_1_ in r band				
168–173	F6.3	---	R2-g	[-0.96/1] R-
square in g band				
175–180	F6.3	---	R2-r	[-0.05/1] R-
square in r band				
182–186	F5.3	mag	gAmp	[0.006/8.11]?
=0 Amplitude in g band				
188–192	F5.3	mag	rAmp	[0.01/7]?=0

Amplitude in r band				
194-201	F8.3	[-]	log(FAP-g)	[-226.2/0]
False alarm probability in g band				
203-210	F8.3	[-]	log(FAP-r)	[-223.2/0]
False alarm probability in r band				
212-216	A5	---	Type	Variable type
(1)				
218-222	F5.3	mag	Dmin-g	[0/2.5]
Minimum difference in g band				
224-228	F5.3	mag	Dmin-r	[0/2.6]
Minimum difference in r band				

Note (1): Variable type as follows:

BYDra = BY Draconis rotational variable (84697 occurrences)

CEP = Classical Cepheid (1262 occurrences)

CEPII = Type II Cepheid (358 occurrences)

DSCT = Low-amplitude {delta} Scuti star (16709 occurrences)

EA = EA-type eclipsing binary (49943 occurrences)

EW = EW-type eclipsing binary (369707 occurrences)

Mira = Mira variable (11879 occurrences)

RR = RRab Lyrae (32518 occurrences)

RRc = RRc Lyrae (13875 occurrences)

RSCVN = RS CVn (Canum Venaticorum) rotational variable (81393 occurrences)

SR = Semiregular variable (119261 occurrences)

Byte-by-byte Description of file: table3.dat

Bytes	Format	Units	Label	Explanations
1- 22	A22	---	ID	ZTF variables catalog identifier
24- 32	F9.5	deg	RAdeg	Right Ascension in decimal degrees (J2000)
34- 42	F9.5	deg	DEdeg	Declination in decimal degrees (J2000)
44- 49	F6.3	mag	gmag	[11.1/22.1]?=0 Mean Palomar/ZTF g band magnitude
51- 56	F6.3	mag	rmag	[11.2/21.6]?=0 Mean Palomar/ZTF r band magnitude
58- 68	F11.7	d	Per-g	[0.02/969]?=0 Period determined in g band
70- 80	F11.7	d	Per-r	[0.02/956]?=0 Period determined in r band
82- 86	F5.3	mag	gAmp	[0.01/6]?=0 Amplitude in g band
88- 92	F5.3	mag	rAmp	[0.008/5.7]?=0 Amplitude in r band
94- 97	I4	---	Ng	[0/1220] Number of detections in g band
99-102	I4	---	Nr	[0/1668] Number of detections in r band
104-111	F8.3	[-]	log(FAP-g)	[-124.4/0] False alarm probability in g band
113-120	F8.3	[-]	log(FAP-r)	[-115.5/0] False alarm probability in r band

Byte-by-byte Description of file: ztf2[gr].dat

Bytes	Format	Units	Label	Explanations
1– 6	I6	---	Seq	[1/781604]
Running sequence number				
8–18	E11.7	deg	RAdeg	Right ascension
in decimal degrees (J2000)				
20–30	E11.7	deg	DEdeg	[-20.8/80.3]
Declination (J2000)				
32–44	F13.7	d	MJD	
[58197.15/58725.2] Modified Julian Date				
46–55	F10.7	mag	mag	[9.97/22.63]
Magnitude in g or r-band				
57–70	E14.7	mag	e_mag	[-0.005/0.4]
Uncertainty on the mag				
72–75	I4	---	Flag	[0/4628] Flag

History:

From electronic version of the journal for
tables

Light curves downloaded from
doi:10.5281/zenodo.3886372 (csv files)

(End)

Prepared by [AAS],

